

ENV 797 - Time Series Analysis for Energy and Environment Applications | Spring 2026

Assignment 4 - Due date 02/10/26

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Directions

You should open the .rmd file corresponding to this assignment on RStudio. The file is available on our class repository on Github. And to do so you will need to fork our repository and link it to your RStudio.

Once you have the file open on your local machine the first thing you will do is rename the file such that it includes your first and last name (e.g., “LuanaLima_TSA_A04_Sp26.Rmd”). Then change “Student Name” on line 4 with your name.

Then you will start working through the assignment by **creating code and output** that answer each question. Be sure to use this assignment document. Your report should contain the answer to each question and any plots/tables you obtained (when applicable).

When you have completed the assignment, **Knit** the text and code into a single PDF file. Submit this pdf using Sakai.

R packages needed for this assignment: “xlsx” or “readxl”, “ggplot2”, “forecast”, “tseries”, and “Kendall”. Install these packages, if you haven’t done yet. Do not forget to load them before running your script, since they are NOT default packages.\

```
rm(list = ls())
```

```
#Load/install required package here
library(tseries)
library(dplyr)
library(forecast)
library(readxl)
library(openxlsx)
library(ggplot2)
library(base)
library(trend)
library(Kendall)
library(lubridate)
```

Questions

Consider the same data you used for A3 from the spreadsheet “Table_10.1_Renewable_Energy_Production_and_Consumption”. The data comes from the US Energy Information and Administration and corresponds to the December 2025 Monthly Energy Review. **For this assignment you will work only with the column “Total Renewable Energy Production”.**

```

#Importing data set - you may copy your code from A3

#Importing data set without change the original file using read.xlsx
energy_data1 <- read_excel(path="./Data/Table_10.1_Renewable_Energy_Production_and_Consumption_by_Source.xlsx")

#Now let's extract the column names from row 11
read_col_names <- read_excel(path="./Data/Table_10.1_Renewable_Energy_Production_and_Consumption_by_Source.xlsx", sheet=11)
colnames(energy_data1) <- read_col_names

#variables I want
selected_vars <- energy_data1[,c("Month", "Total Renewable Energy Production")]

#turn into a dataframe
new_df <- as.data.frame(selected_vars)
head(new_df)

```

```

##           Month Total Renewable Energy Production
## 1 1973-01-01                      219.839
## 2 1973-02-01                      197.330
## 3 1973-03-01                      218.686
## 4 1973-04-01                      209.330
## 5 1973-05-01                      215.982
## 6 1973-06-01                      208.249

```

```

#make original data into a time series
tsdata <- ts(new_df[,1], frequency = 12, start=c(1973,1))

```

Stochastic Trend and Stationarity Tests

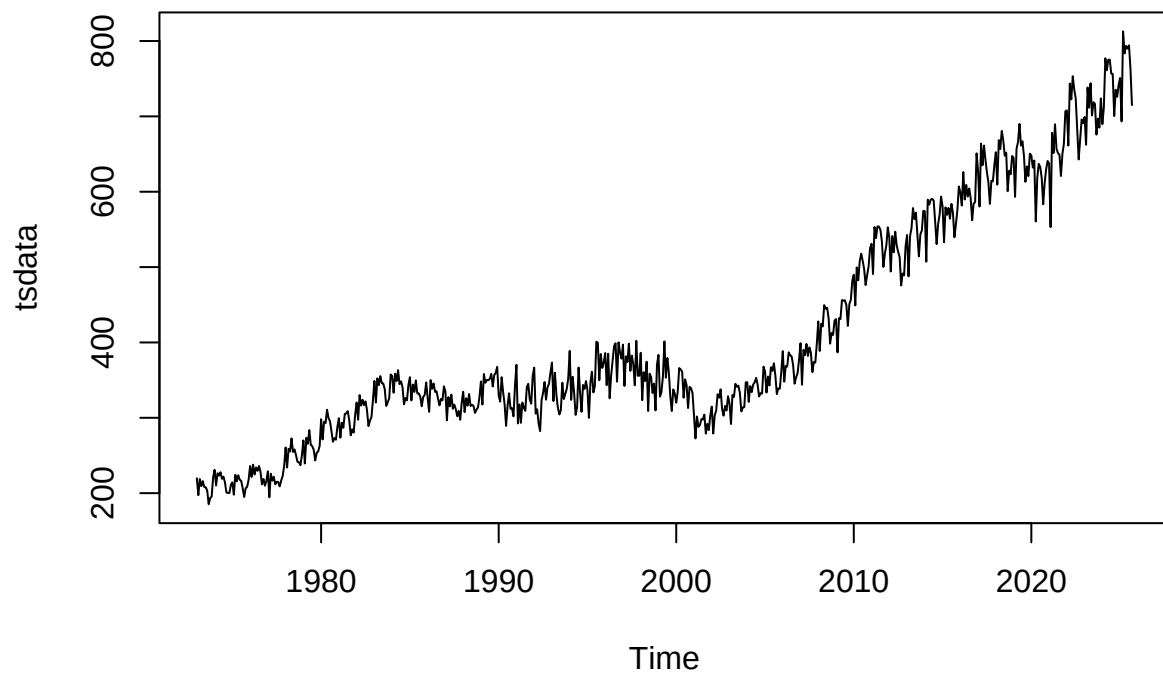
For this part you will work only with the column Total Renewable Energy Production.

Q1

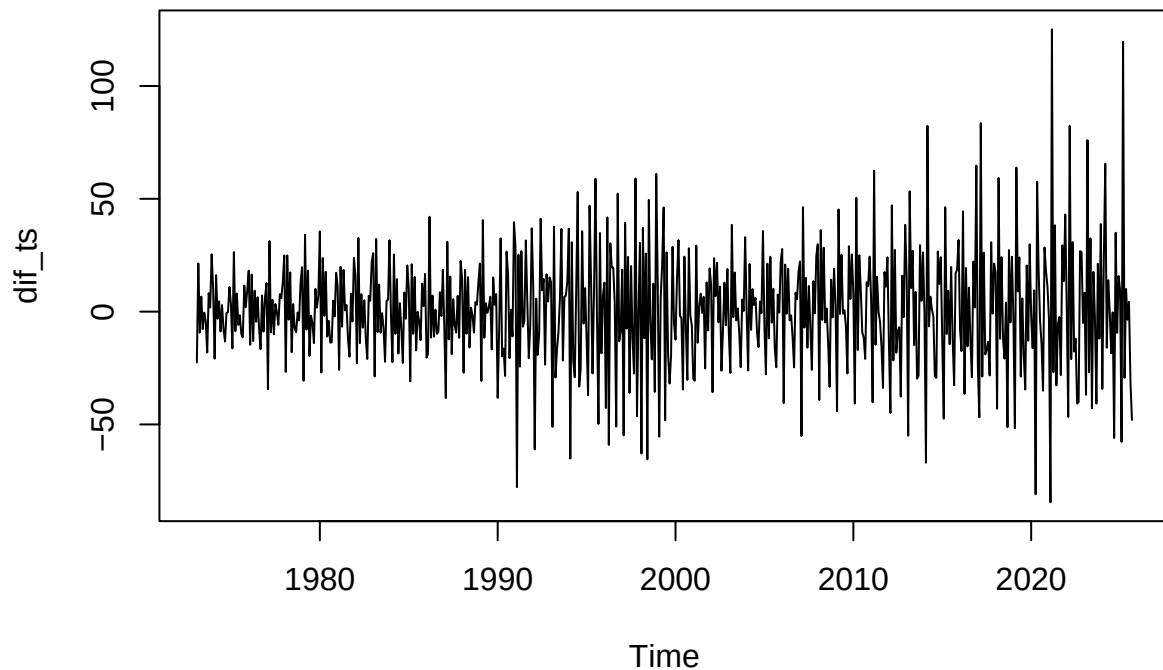
Difference the “Total Renewable Energy Production” series using function `diff()`. Function `diff()` is from package `base` and take three main arguments: * *x* vector containing values to be differenced; * *lag* integer indicating with lag to use; * *differences* integer indicating how many times series should be differenced.

Try differencing at lag 1 only once, i.e., make `lag=1` and `differences=1`. Plot the differenced series. Do the series still seem to have trend?

```
plot(tsdata)
```



```
dif_ts <- diff(tsdata, lag=1, differences=1)
#plot the differenced series
plot(dif_ts)
```



The trend appears to be removed. Now the series is flat around 0 but still spiking in some manner, maybe seasonally. The series does still seem to grow over time.

Q2

Copy and paste part of your code for A3 where you run the regression for Total Renewable Energy Production and subtract that from the original series. This should be the code for Q3 and Q4. make sure you use assign same name for the time series object that you had in A3, otherwise the code will not work.

```
nobs <- nrow(energy_data1)
#Create vector t
t<- c(1:nobs)

#Fit trend for Renewable Energy
RE_linear_trend <- lm(new_df[,2]~t)# plus one bc of date column in the data frame
summary(RE_linear_trend)
```

```
##
## Call:
## lm(formula = new_df[, 2] ~ t)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -154.81  -39.55   12.52   41.49  171.15
##
```

```
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) 171.44868    5.11085   33.55  <2e-16 ***
## t           0.74999     0.01397   53.69  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 64.22 on 631 degrees of freedom
## Multiple R-squared:  0.8204, Adjusted R-squared:  0.8201
## F-statistic: 2883 on 1 and 631 DF, p-value: < 2.2e-16
```

```
#store coefficients
RE_beta0 <- as.numeric(RE_linear_trend$coefficients[1]) # intercept
RE_beta1 <- as.numeric(RE_linear_trend$coefficients[2]) # slope

detrend_RE <- new_df[,2] - (RE_beta0 + RE_beta1*t)
#transform to ts
ts_detrend_RE <- ts(detrend_RE,frequency = 12, start=c(1973,1))
```

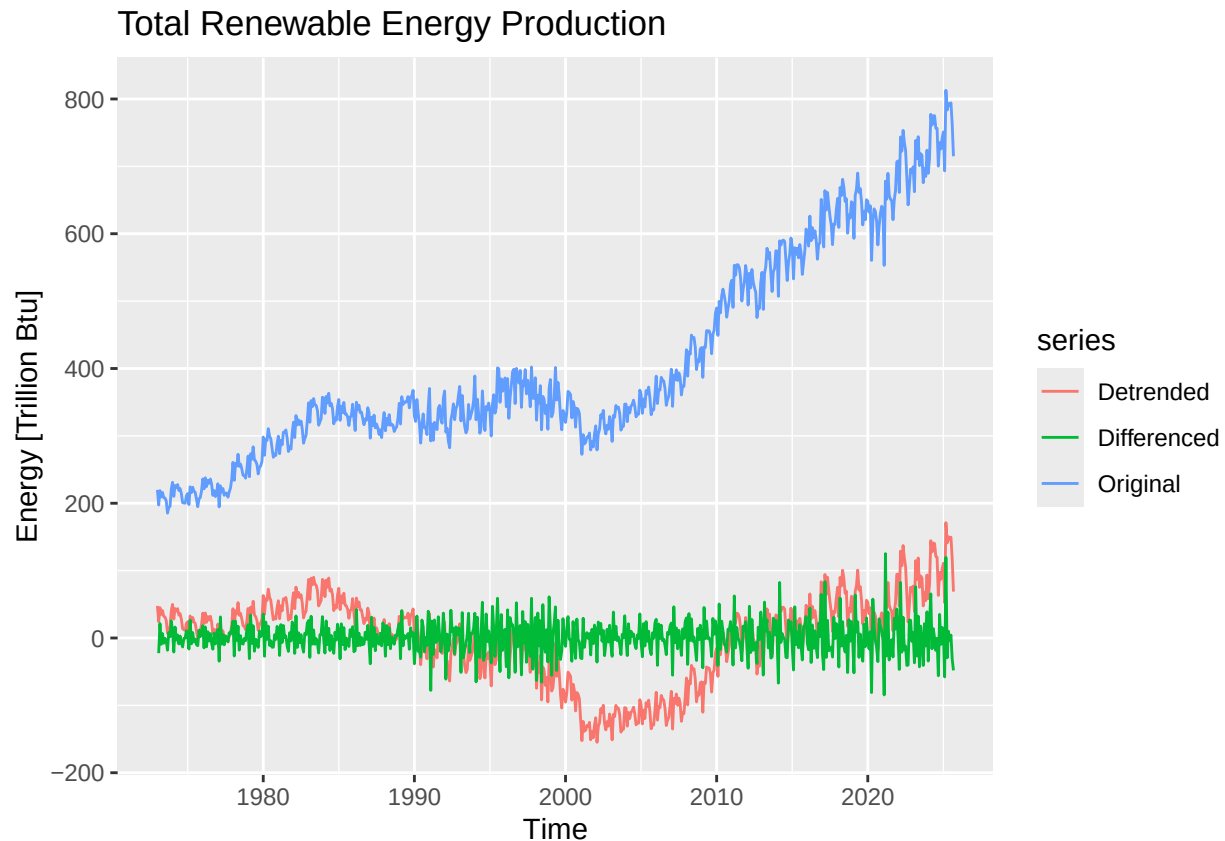
Q3

Now let's compare the differenced series with the detrended series you calculated on A3. In other words, for the "Total Renewable Energy Production" compare the differenced series from Q1 with the series you detrended in Q2 using linear regression.

Using `autoplot()` + `autolayer()` create a plot that shows the three series together (i.e. "Original", "Differenced", "Detrended `lm()`"). Make sure your plot has a legend. The easiest way to do it is by adding the `series=` argument to each `autoplot` and `autolayer` function. Look at the key for A03 for an example on how to use `autoplot()` and `autolayer()`.

What can you tell from this plot? Which method seems to have been more efficient in removing the trend?

```
autoplot(tsdata,series="Original") +
  autolayer(ts_detrend_RE,series="Detrended") +
  autolayer(dif_ts, series="Differenced") +
  ylab("Energy [Trillion Btu]") +
  ggtitle("Total Renewable Energy Production")
```



Answer: The trend in the differenced data seems far more reduced than the detrended data. Detrending the data seemed to tilt but not actually remove the trend, whereas the differenced data does not have the same trend as the original or detrended data exhibit.

Q4

Plot the ACF for the three series and compare the plots. Add the argument `ylim=c(-0.5,1)` to the `autoplot()` or `Acf()` function - whichever you are using to generate the plots - to make sure all three y axis have the same limits. Looking at the ACF which method do you think was more efficient in eliminating the trend? The linear regression or differencing?

```
library(cowplot)
```

```
##
## Attaching package: 'cowplot'

## The following object is masked from 'package:lubridate':
##
## stamp
```

```
plot_grid(
  autoplot(Acf(tsddata, lag.max=40, plot=FALSE),
    main="Renewable Original",
    ylim=c(-0.5,1)), #ylim is used to set the y axis limits
```

```

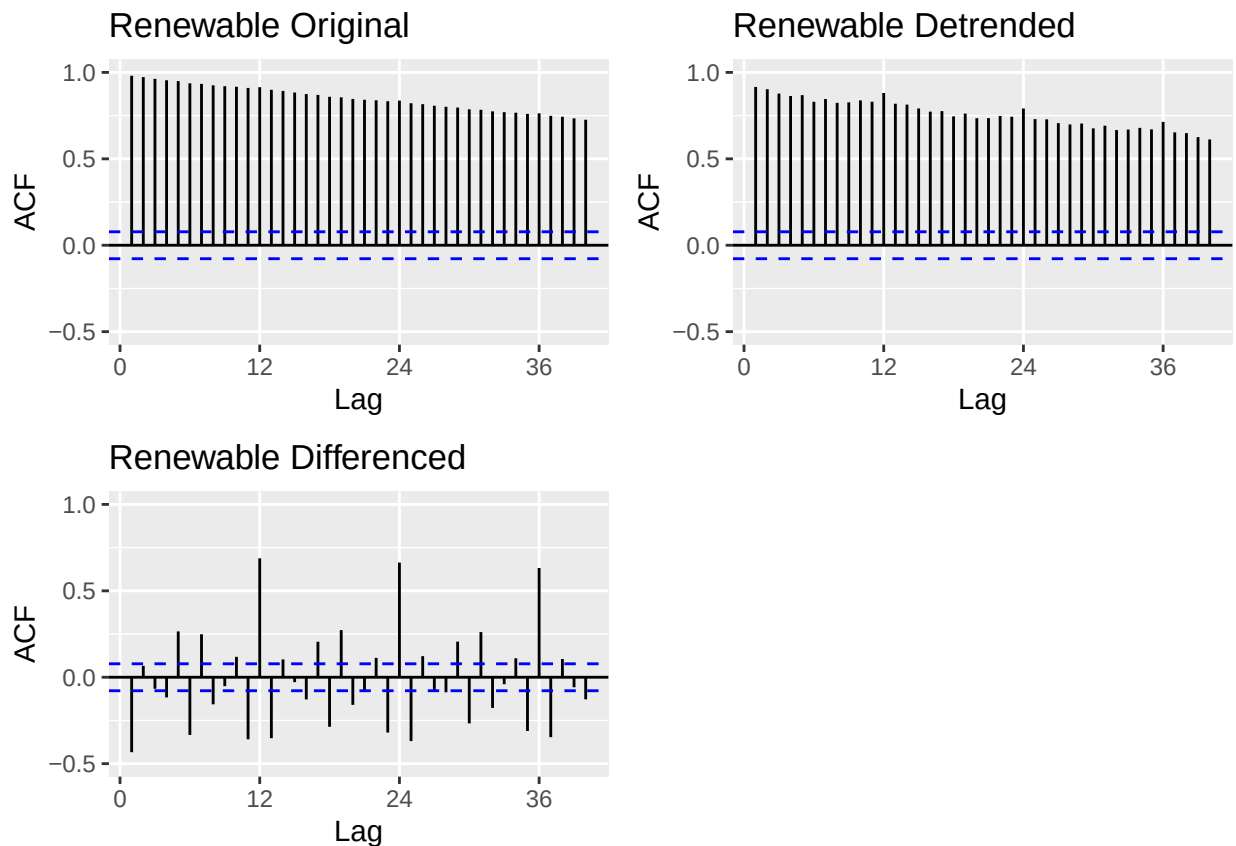
autoplot(Acf(ts_detrend_RE,lag.max=40,plot=FALSE),
  main="Renewable Detrended",
  ylim=c(-0.5,1)),
autoplot(Acf(dif_ts,lag.max=40,plot=FALSE),
  main="Renewable Differenced",
  ylim=c(-0.5,1))
)

```

```

## Warning in ggplot2::geom_segment(lineend = "butt", ...): Ignoring unknown parameters: 'main' and 'ylim'
## Ignoring unknown parameters: 'main' and 'ylim'
## Ignoring unknown parameters: 'main' and 'ylim'

```



Answer: Looking at the three plots, it looks like differencing did the best at eliminating the trend. The linear regression is barely different than the original series ACF, the trend is not removed.

Q5

Compute the Seasonal Mann-Kendall and ADF Test for the original “Total Renewable Energy Production” series. Ask R to print the results. Interpret the results for both test. What is the conclusion from the Seasonal Mann Kendall test? What’s the conclusion for the ADF test? Do they match what you observed in Q3 plot? Recall that having a unit root means the series has a stochastic trend. And when a series has stochastic trend we need to use differencing to remove the trend.

```
#Seasonal Mann-Kendall
SMKtest <- SeasonalMannKendall(tsddata)
print("Results for Seasonal Mann Kendall /n")
```

```
## [1] "Results for Seasonal Mann Kendall /n"
```

```
print(summary(SMKtest))
```

```
## Score = 13083 , Var(Score) = 201135
## denominator = 16379.5
## tau = 0.799, 2-sided pvalue =< 2.22e-16
## NULL
```

```
#ADF Test
#Null hypothesis is that data has a unit root
print("Results for ADF test/n")
```

```
## [1] "Results for ADF test/n"
```

```
print(adf.test(tsddata,alternative = "stationary"))
```

```
##
## Augmented Dickey-Fuller Test
##
## data: tsdata
## Dickey-Fuller = -1.0247, Lag order = 8, p-value = 0.9347
## alternative hypothesis: stationary
```

Answer: The Mann-Kendall test result provides strong evidence to reject the null hypothesis that there is no trend, and support the alternative hypothesis that there is a trend in the original data. The ADF test has a high p-value of 0.9347, this means I cannot reject the null hypothesis that the data has a unit root and is non-stationary. This means that the series likely has a stochastic trend. This matches my plots from Q3 - where I needed to difference to remove the trend.

Q6

Aggregate the original “Total Renewable Energy Production” series by year. You can use the same procedure we used in class. Store series in a matrix where rows represent months and columns represent years. And then take the columns mean using function `colMeans()`. Recall the goal is the remove the seasonal variation from the series to check for trend. Convert the accumulates yearly series into a time series object and plot the series using `autoplot()`.

```
#Group data in yearly steps instances
data_matrix <- matrix(tsddata,byrow=FALSE,nrow=12)
```

```
## Warning in matrix(tsddata, byrow = FALSE, nrow = 12): data length [633] is not a
## sub-multiple or multiple of the number of rows [12]
```



```

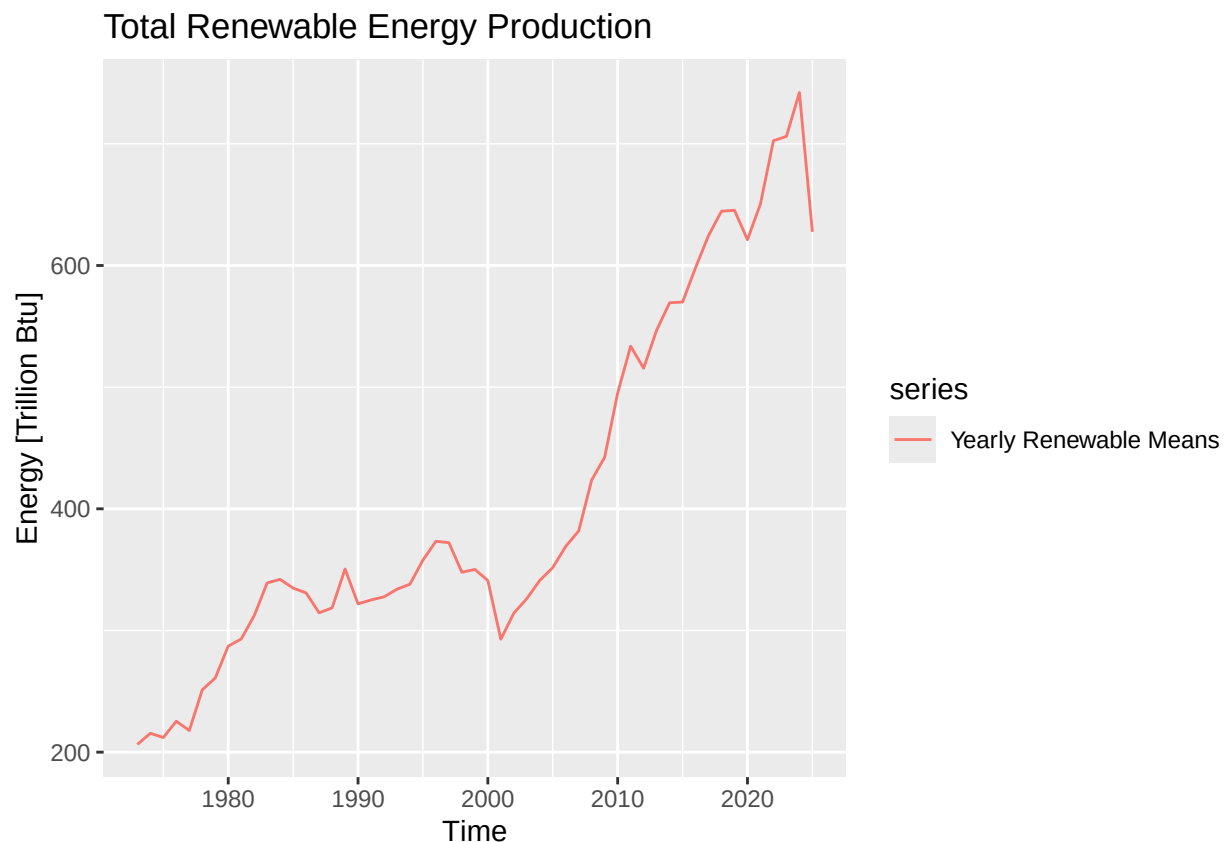
means_yearly <- colMeans(data_matrix)

my_year <- c(year(first(new_df$Month)):year(last(new_df$Month)))
#data frame with yearly means
data_yearly <- data.frame(my_year, means_yearly)

#create a time series of yearly means, note to self: the -1 is removing the garbage column creating a t
yearly_ts <- ts(data_yearly[,-1],frequency = 1, start=c(1973))

autoplot(yearly_ts,series="Yearly Renewable Means") +
  ylab("Energy [Trillion Btu]") +
  ggtitle("Total Renewable Energy Production")

```



Q7

Apply the Mann Kendall, Spearman correlation rank test and ADF. Are the results from the test in agreement with the test results for the monthly series, i.e., results for Q5?

```

#Seasonal Mann-Kendall
SMKtest2 <- SeasonalMannKendall(yearly_ts)
print("Results for Seasonal Mann Kendall /n")

```

```
## [1] "Results for Seasonal Mann Kendall /n"
```

```
print(summary(SMKtest2))
```

```
## Score = 1124 , Var(Score) = 16995.33
## denominator = 1378
## tau = 0.816, 2-sided pvalue =< 2.22e-16
## NULL
```

p value of <0.05 by far

```
#Spearman Correlation Test
print("Results from Spearman Correlation")
```

```
## [1] "Results from Spearman Correlation"
```

```
sp_rho=cor.test(yearly_ts,my_year,method="spearman")
print(sp_rho)
```

```
##
## Spearman's rank correlation rho
##
## data: yearly_ts and my_year
## S = 1898, p-value < 2.2e-16
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##      rho
## 0.9234801
```

same p value result as mann-kendall

```
#ADF Test
#Null hypothesis is that data has a unit root
print("Results for ADF test/n")
```

```
## [1] "Results for ADF test/n"
```

```
print(adf.test(yearly_ts,alternative = "stationary"))
```

```
##
## Augmented Dickey-Fuller Test
##
## data: yearly_ts
## Dickey-Fuller = -1.6789, Lag order = 3, p-value = 0.7037
## alternative hypothesis: stationary
```

p value = 0.7037 > Answer: Yes, both spearman and mann-kendall provide evidence to reject the null, indicating there is a trend and the p-value in the ADF test is still >0.05 indicating there is a unit root/stochastic trend. This does match my Q5 answers still.